

Knowledge Distillation Across Teacher Architectures for Edge-Efficient Plant Disease Classification

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Abstract. Accurate image-based plant disease classification can support precision agriculture, but high-capacity networks are not always convenient for edge or mobile deployment. This work studies knowledge distillation from large teacher networks into a compact MobileNetV3-Small student, and asks in particular whether the *architecture of the teacher* affects the distilled student. As a preliminary demonstration, we distill a ResNet50 teacher into MobileNetV3-Small on the 38-class PlantVillage dataset with a single train/validation/test split; the distilled student reaches 0.9937 accuracy and 0.9907 macro-F1 with 1.56M parameters, about 15 times fewer than the teacher. We then carry out the main study on the smaller and class-imbalanced Cassava leaf-disease dataset (five classes) using stratified five-fold cross-validation. Three teachers from different architectural families --- ResNet50 (residual CNN), EfficientNet-B2 (compound-scaled CNN), and ViT-B/16 (vision transformer) --- are each retrained per fold and distilled into the same student, and we report accuracy together with macro-averaged precision, recall, and F1. The cross-validated comparison provides a more statistically grounded view of whether distillation helps on a harder dataset and of how much the teacher architecture matters.

Keywords: Knowledge distillation · Plant disease classification · MobileNetV3 · Cross-validation · Vision transformer · Edge inference.

1 Introduction

Plant disease recognition is a practical computer vision problem with direct relevance to crop monitoring and precision agriculture. Deep neural networks can achieve strong classification accuracy on curated leaf image datasets, but models with high parameter counts may be inconvenient for mobile or edge-oriented deployment. This creates a tension between predictive accuracy and computational efficiency.

This work studies that trade-off through knowledge distillation, in which a high-capacity teacher network guides the training of a lightweight MobileNetV3-Small student. The aim is not to claim deployment on a specific device, but to evaluate whether a compact student can retain a teacher’s predictive behavior

while using far fewer parameters. Beyond the standard teacher-to-student comparison, we focus on a question that a single experiment cannot answer: does the *choice of teacher architecture* change how well the student learns? To study this we compare three teachers from distinct design families under a cross-validated protocol.

The contributions are as follows. First, as a preliminary demonstration, we distill a ResNet50 teacher into MobileNetV3-Small on the 38-class PlantVillage dataset using a single split. Second, and as the main study, we evaluate three teacher architectures --- ResNet50, EfficientNet-B2, and ViT-B/16 --- distilled into the same student on the Cassava leaf-disease dataset. Third, we use a methodologically careful protocol: stratified five-fold cross-validation with the teacher retrained from scratch on each fold’s training partition, so that the held-out fold never influences the teacher. Fourth, we report macro-averaged precision and recall alongside accuracy and macro-F1, which is important given the class imbalance of the Cassava data, and present results as mean $\pm$ standard deviation across folds.

2 Background and Related Work

PlantVillage provides a widely used collection of labeled plant leaf images for plant health research [8]. The dataset has enabled controlled studies of image-based plant disease classification, although its curated imaging conditions are cleaner than many field scenarios. The Cassava leaf-disease data used in the main study [12] consists of five classes (four disease categories and a healthy class) and is more class-imbalanced, providing a harder and less controlled testbed. These two datasets were chosen deliberately: PlantVillage is a clean, widely used benchmark that is well suited to an initial controlled demonstration of the distillation pipeline, whereas Cassava consists of real competition field imagery with strong class imbalance, so it tests whether the same conclusions hold under more realistic and challenging conditions. Both are also directly relevant to the edge-deployment motivation, since plant disease diagnosis is a natural use case for on-device inference in the field.

Deep learning has been applied extensively to plant disease classification. Early work showed that convolutional networks trained on PlantVillage can classify leaf diseases with high accuracy [10], and later studies compared fine-tuning strategies across architectures for this task [14]. Most of this literature focuses on maximizing accuracy with large backbones; comparatively little attention has been paid to compressing these models for on-device inference, which is the gap this work addresses through distillation.

Knowledge distillation was introduced as a way to transfer information from a larger teacher model to a smaller student model [6]. Instead of learning only from one-hot labels, the student also learns from the teacher’s softened class probabilities, which can encode similarities between classes, making distillation attractive when the deployment model must be smaller than the model used to generate the strongest predictions. Beyond matching output logits, later methods

transfer intermediate representations, such as hidden-layer hints [11] and spatial attention maps [16]; a broad overview of these variants is given by Gou et al. [4]. In this work we use the original logit-based formulation, since our focus is the choice of teacher rather than the distillation mechanism.

A recurring and directly relevant finding is that a more accurate teacher does not necessarily produce a better student. Cho and Hariharan [2] show that very large teachers can distill less effectively into small students because of the capacity gap between them, and Mirzadeh et al. [9] introduce intermediate “teacher-assistant” networks to bridge that gap. Beyer et al. [1] further argue that the consistency of the teacher-student training setup matters more than raw teacher accuracy. These results motivate our central question: when the student is fixed, does the teacher’s *architecture* --- and not merely its accuracy --- affect the distilled student? Distillation has also been studied in the context of vision transformers, most notably DeiT [15], which distills *into* a transformer using a CNN teacher. Our setting is complementary: we distill *from* a transformer (and from two CNNs) into a compact convolutional student, comparing teacher families under a single protocol.

The teachers in this work span three architectural families. ResNet50 is a residual network that uses skip connections to ease optimization of deeper convolutional models [5]. EfficientNet-B2 is a convolutional network obtained by compound scaling of depth, width, and resolution, and is known for strong accuracy at a low parameter count [13]. ViT-B/16 is a vision transformer that treats an image as a sequence of patches and relies on self-attention rather than convolution [3], giving it a substantially different inductive bias from the two CNN teachers. The student in all experiments is MobileNetV3-Small, an architecture designed for mobile inference through efficient convolutional blocks and hardware-aware design choices [7]. These models were chosen for a specific reason. The three teachers are well-established, strong image classifiers with pretrained ImageNet weights, but they represent deliberately different design philosophies --- residual convolutions, compound-scaled convolutions, and patch-based self-attention --- and they also differ widely in parameter count, which lets us separate the effect of teacher architecture from that of teacher size. MobileNetV3-Small was chosen as the single student because it is a representative compact model intended for edge and mobile deployment, making it the natural target for compression. Comparing CNN and transformer teachers that distill into this same convolutional student lets us probe whether teacher architecture, and not only teacher accuracy, influences the student.

3 Methodology

Let z_t and z_s denote the teacher and student logits for an input image, and let y denote the ground-truth class label. Supervised teacher and supervised student baselines are trained with cross-entropy against the hard labels. For distillation, the teacher is first trained and then frozen, and the student is optimized using a

weighted sum of the cross-entropy loss and a temperature-scaled Kullback--Leibler divergence term:

$$\mathcal{L} = \alpha T^2 \text{KL} \left(\text{softmax} \left(\frac{z_t}{T} \right) \parallel \text{softmax} \left(\frac{z_s}{T} \right) \right) + (1 - \alpha) \text{CE}(y, z_s). \quad (1)$$

The T^2 factor follows the common practice of compensating for the gradient scaling introduced by softened probabilities. The temperature is fixed at $T = 4.0$ throughout. The distillation weight is $\alpha = 0.5$ in the preliminary PlantVillage experiment and $\alpha = 0.7$ in the main Cassava study; in both cases α is held constant across teachers so that any difference between teachers is attributable to the teacher rather than to the loss weighting. MobileNetV3-Small is used as the student in every experiment because it has substantially fewer parameters than any of the teachers while retaining a convolutional design appropriate for image classification.

4 Experimental Setup

All experiments use pretrained ImageNet weights, images resized to 224×224 pixels and normalized with ImageNet statistics, random horizontal flipping and small random rotations during training, deterministic resizing for validation and test, and the best validation checkpoint for final evaluation. Inference latency is reported from a batch-one timing routine that averages repeated forward passes; latency values are local hardware measurements rather than hardware-independent deployment benchmarks.

4.1 Main Study: Cross-Validated Multi-Teacher Distillation on Cassava

The main study uses the Cassava leaf-disease dataset (five classes) and stratified five-fold cross-validation. The dataset is partitioned into five stratified folds, preserving class proportions. For each fold, that fold serves as the test set, and the remaining data is split (again with stratification) into training and validation. Crucially, the teacher is *retrained from scratch on that fold’s training partition* before being frozen and used to distill the student. This avoids the optimistic bias that would arise if a single teacher had been exposed to images that later appear in a fold’s test set. The fold partition is fixed by seed, so every teacher and every model sees identical folds. All models are initialized from ImageNet-pretrained weights, so the comparison concerns transfer to the plant-disease task rather than training from random initialization; this pretraining is shared across all teachers and the student and therefore does not bias the teacher comparison.

Three teachers are evaluated --- ResNet50, EfficientNet-B2, and ViT-B/16 --- each distilled into a MobileNetV3-Small student, and compared against a supervised MobileNetV3-Small baseline. The student schedule and distillation hyperparameters ($T = 4.0$, $\alpha = 0.7$) are held constant across teachers. Per-fold

teacher schedules and the student schedule are summarized in Table 1. Each teacher is also evaluated on its own, trained supervised under the same cross-validation, so that the teachers’ own cross-validated performance can be reported alongside the students. These runs use an NVIDIA RTX 4070 (CUDA) with automatic mixed precision. For each model we report the mean and standard deviation across the five folds of accuracy and of macro-averaged precision, recall, and F1; macro averaging weights each class equally, which matters because the Cassava classes are imbalanced. Because the folds are shared across models, each distilled student can be compared against the supervised baseline with a paired two-sided t -test on the five per-fold accuracies; we report the resulting p -values so that differences are not over-interpreted given the small number of folds.

Table 1. Per-fold training configuration for the Cassava cross-validation study. All teachers are distilled into the same MobileNetV3-Small student with $T = 4.0$ and $\alpha = 0.7$.

Model	Role	Learning rate	Epochs	Batch
ResNet50	Teacher	3×10^{-4}	10	64
EfficientNet-B2	Teacher	3×10^{-4}	10	64
ViT-B/16	Teacher	1×10^{-4}	10	32
MobileNetV3-Small	Student / baseline	5×10^{-4}	15	64

4.2 Preliminary Study: Single-Split Distillation on PlantVillage

As a preliminary demonstration, a ResNet50 teacher, a supervised MobileNetV3-Small baseline, and a distilled MobileNetV3-Small student are trained on the full 38-class PlantVillage dataset with a single deterministic 70/15/15 train/validation/test split (seed 42). These runs use 15 epochs, weight decay 10^{-4} , batch size 32, and distillation settings $T = 4.0$ and $\alpha = 0.5$, and were executed on Apple Silicon (MPS). The ResNet50 teacher uses learning rate 3×10^{-4} , the supervised student uses 10^{-3} , and the distilled student uses 5×10^{-4} .

5 Results and Discussion

5.1 Main Study: Cassava Cross-Validation

Table 2 summarizes the cross-validated results on Cassava. The table reports each teacher trained supervised, the supervised MobileNetV3-Small baseline (teacher “--”), and the student distilled from each teacher, as mean $\pm$ standard deviation across the five folds. Two questions are of interest: whether distillation improves the student over the supervised baseline on this harder, imbalanced dataset, and whether a stronger or architecturally different teacher distills better than ResNet50.

Table 2. Cross-validated test results on cassava (5-fold, mean $\pm$ standard deviation across folds). Precision, recall, and F1 are macro-averaged. The teacher is retrained per fold; the student and KD hyperparameters are fixed.

Teacher	Student	Accuracy (%)	Precision (%)	Recall (%)	Macro-F1 (%)	Parameters
--	mobilenet_v3_small	--	--	--	--	--
resnet50	mobilenet_v3_small	--	--	--	--	--
efficientnet_b2	mobilenet_v3_small	--	--	--	--	--
vit_b_16	mobilenet_v3_small	--	--	--	--	--

Reporting mean $\pm$ standard deviation across folds, rather than a single split, makes these comparisons more robust to the particular partition of the data. Because precision and recall are reported alongside accuracy, the table also exposes whether any accuracy gains are accompanied by balanced per-class behavior rather than improvements only on the majority class.

5.2 Sensitivity to Distillation Hyperparameters

The main study fixes the distillation temperature and weight at the widely used values $T = 4.0$ and $\alpha = 0.7$, and holds them constant across teachers so that differences are attributable to the teacher. To check that this choice is reasonable and that the comparison is not an artifact of a particular setting, we run a single-split sensitivity analysis for the ResNet50 teacher on fold 0 of the Cassava partition. Because T and α affect only the student loss, the teacher is trained once and reused for every setting; we then sweep $T \in \{1, 2, 4, 8\}$ at $\alpha = 0.7$ and $\alpha \in \{0.3, 0.5, 0.7, 0.9\}$ at $T = 4.0$. Table 3 reports the resulting student accuracy and macro-F1.

Table 3. Single-split KD sensitivity analysis for the resnet50 teacher (fold 0 of the Cassava cross-validation split). Temperature is swept at $\alpha = 0.7$ and α at $T = 4$; † marks the default used in the main study. Values are single-split point estimates (no fold error bars).

Sweep	T	α	Accuracy (%)	Macro-F1 (%)
temperature 1	0.7		--	--
temperature 2	0.7		--	--
temperature 4	0.7 [†]		--	--
temperature 8	0.7		--	--
alpha	4	0.3	--	--
alpha	4	0.5	--	--
alpha	4	0.7 [†]	--	--
alpha	4	0.9	--	--

These results are single-split point estimates without fold error bars, so they are intended as a sensitivity check rather than as a second cross-validated comparison.

5.3 Preliminary Study: PlantVillage Single Split

Table 4 reports the preliminary single-split results on the 38-class PlantVillage task. The ResNet50 teacher reaches 0.9936 test accuracy and 0.9912 macro-F1. The supervised MobileNetV3-Small baseline is already strong, with 0.9851 accuracy and 0.9779 macro-F1. After distillation from ResNet50, MobileNetV3-Small improves to 0.9937 accuracy and 0.9907 macro-F1 while retaining the same 1.56M parameter count.

Table 4. Preliminary single-split test results on the 38-class PlantVillage task.

Model	Training strategy	Accuracy	Macro-F1	Parameters	Latency
ResNet50	Teacher supervised	0.9936	0.9912	23,585,894	5.47 ms
MobileNetV3-Small	Supervised baseline	0.9851	0.9779	1,556,806	3.51 ms
MobileNetV3-Small	Distilled from ResNet50	0.9937	0.9907	1,556,806	3.53 ms

The comparison between the supervised and distilled MobileNetV3-Small models is informative because the architecture and parameter count are identical in both rows. The improvement from 0.9851 to 0.9937 accuracy therefore does not come from increasing student capacity; it indicates that the teacher logits provide useful training signal beyond the hard labels alone. At the same time, PlantVillage images are relatively clean and the supervised student is already very strong, which limits the headroom for distillation. This motivates the main cross-validated study on the harder Cassava data, where the supervised baseline is weaker and the effect of the teacher can be observed more clearly.

6 Reproducibility and Limitations

All experiments are configuration-driven. The preliminary PlantVillage runs correspond to the full-dataset teacher, student, and distillation configurations, while the main study is driven by the cross-validation configs and executed with the `src.cross_validate` runner; results are collated into a single table with `src.collate_cv`. Reported metrics are taken from the stored result files rather than copied from terminal logs, so the report stays tied to reproducible artifacts: each run stores a model summary, per-fold metrics, final aggregated metrics, and result CSVs.

Several limitations remain. First, the distillation hyperparameters are kept fixed across teachers ($T = 4.0$, and $\alpha = 0.5$ or 0.7); a more complete study would tune temperature and loss weighting per teacher, which would help separate the

effect of the teacher architecture from that of the distillation schedule. Second, the per-fold teacher retraining means each teacher is trained multiple times, which is computationally expensive but necessary to avoid test leakage. Third, latency is measured locally --- on Apple Silicon (MPS) for the preliminary study and on an NVIDIA RTX 4070 (CUDA) for the main study --- so latency values are comparable only within each study and should not be read as deployment benchmarks. Finally, both datasets are curated; performance on field images with cluttered backgrounds, varying lighting, and different cameras may be lower.

7 Conclusion

This work studied knowledge distillation from large teachers into a compact MobileNetV3-Small student for edge-efficient plant disease classification. A preliminary single-split experiment on the 38-class PlantVillage dataset showed that a ResNet50 teacher can be distilled into a student with 0.9937 accuracy and 0.9907 macro-F1 at about 15 times fewer parameters than the teacher. The main study then used stratified five-fold cross-validation on the harder, imbalanced Cassava dataset to compare three teacher architectures --- ResNet50, EfficientNet-B2, and ViT-B/16 --- distilled into the same student, reporting accuracy together with macro-averaged precision, recall, and F1. The main limitations are dataset realism and the use of fixed distillation hyperparameters; future work should tune the distillation schedule per teacher, evaluate on more realistic field imagery, and measure latency on actual edge hardware.

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